The Validity of Kalman Filtering for Cross-Asset Return Prediction

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Abstract

This paper evaluates Kalman filter-based state-space models for predicting PepsiCo (PEP) returns using Coca-Cola (KO) as an exogenous predictor over the period 2016–2022. The Kalman filter is widely used for estimating hidden, time-varying relationships in noisy systems. However, financial markets violate key assumptions underlying this framework, including smooth evolution, Gaussian noise, and stable relationships over time.

We show that while adaptive Kalman models are theoretically designed to handle changing environments, they can fail in practice due to instability during regime shifts. In particular, models that attempt to react strongly to market changes can amplify noise instead of improving predictions.

Background

Kalman filtering is a mathematical method used to estimate hidden quantities that cannot be directly observed but can only be inferred through noisy measurements.

At a high level, the system assumes two things:

- There is a hidden state (what we want to estimate)
- We observe noisy measurements related to that state

This is written as:

$$x_t = Ax_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q)$$

$$z_t = Hx_t + v_t, \quad v_t \sim \mathcal{N}(0, R)$$

To understand this intuitively:

- x_t: hidden state (true underlying system)

- z_t: observed measurement (what we actually see)
- A: how the system evolves over time
- H: how the hidden state maps to observations
- w_t: process noise (random evolution in the system)
- v_t: measurement noise (errors in observation)

The Kalman filter assumes:

- linear dynamics
- Gaussian noise
- smooth evolution over time

Because of this, it is widely used in systems where changes are gradual, such as aerospace tracking, radar systems, and signal processing.

Motivation

This work comes from my experience applying Kalman filters in computer vision for object tracking. During an engineering design project, I was tasked with creating a product that allowed my client to use the touchscreen in his vehicle in the presence of hand tremors. I leveraged the filter as a smoothing algorithm to reduce the perceived jitter in his hand.

In those systems, the filter performs well when motion is smooth. However, it struggles when assumptions break, such as:

- occlusion (object disappears temporarily)
- sudden motion changes
- missing or corrupted observations

This suggests a key limitation: Kalman filters depend heavily on smooth dynamics.

Financial markets are fundamentally different. They are driven by:

- human behavior
- macroeconomic policy
- external shocks (crashes, pandemics)

These systems often change abruptly rather than smoothly.

Despite this, Kalman filters are widely used in finance to estimate evolving relationships between assets.

This raises the central question:

Can a model built on smooth physical assumptions still work in highly unstable financial systems?

Problem Formulation

We model PepsiCo returns using Coca-Cola returns.

Returns are defined as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

We model:

$$r_t^{PEP} = H_t x_t + v_t$$

where:

- r_t^{PEP}: PepsiCo return at time t
- r_t^{KO}: Coca-Cola return (input signal)
- x_t: hidden time-varying relationship between KO and PEP
- H_t: design matrix built from KO returns
- v_t: noise in financial markets

The hidden relationship evolves as:

$$x_t = x_{t-1} + w_t$$

where w_t represents small random changes in the relationship.

Model Specifications

Model 1: Linear Kalman Filter

We begin by assuming that PEP returns are generated by a linear relationship with lagged KO returns plus additive noise:

$$r_t^{PEP} = \alpha_t + \beta_t r_{t-1}^{KO} + v_t$$

where the measurement noise is assumed Gaussian:

$$v_t \sim \mathcal{N}(0, R)$$

Probabilistic interpretation

The above implies the conditional likelihood:

$$p(r_t^{PEP} \mid \alpha_t, \beta_t) = \mathcal{N}(\alpha_t + \beta_t r_{t-1}^{KO}, R)$$

Over time, assuming independence:

$$p(\mathbf{r}) = \prod_t \mathcal{N}(r_t^{PEP} \mid \alpha_t + \beta_t r_{t-1}^{KO}, R)$$

Log-likelihood and least squares form

Taking the log:

$$\log p(\mathbf{r}) = -\frac{1}{2} \sum_t \frac{\left(r_t^{PEP} - \alpha_t - \beta_t r_{t-1}^{KO}\right)^2}{R} + C$$

Maximizing likelihood is therefore equivalent to minimizing:

$$\sum_t \left(r_t^{PEP} - \alpha_t - \beta_t r_{t-1}^{KO}\right)^2$$

This is a weighted least squares problem (MSE minimization).

Linear algebra form

We rewrite the observation model as:

$$r_t^{PEP} = H_t x_t + v_t$$

where:

$$x_t = \begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} \quad H_t = \begin{bmatrix} 1 & r_{t-1}^{KO} \end{bmatrix}$$

State evolution assumption

To allow time variation in parameters, we assume a random walk:

$$x_t = x_{t-1} + w_t$$

$$w_t \sim \mathcal{N}(0, Q)$$

So: - parameters change slowly over time - uncertainty is captured by Q

Prediction distribution

Taking expectations:

$$\hat{x}_{t|t-1} = \mathbb{E}[x_t \mid t-1] = \hat{x}_{t-1|t-1}$$

Covariance propagation:

$$P_{t|t-1} = P_{t-1|t-1} + Q$$

Predict observation

Predicted return:

$$\hat{r}_t^{PEP} = H_t \hat{x}_{t|t-1}$$

Residual (innovation):

$$e_t = r_t^{PEP} - H_t \hat{x}_{t|t-1}$$

Innovation uncertainty

$$S_t = H_t P_{t|t-1} H_t^T + R$$

Optimal update (minimum MSE)

Kalman gain is chosen to minimize posterior covariance:

$$K_t = P_{t|t-1} H_t^T S_t^{-1}$$

Update state:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t e_t$$

Update covariance:

$$P_{t|t} = (I - K_t H_t) P_{t|t-1}$$

Final regression form

Substituting the state back into the observation equation gives:

$$r_t^{PEP} = \alpha_t + \beta_t r_{t-1}^{KO}$$

Where:

- α_t: baseline return level of PEP
- β_t: sensitivity of PEP to KO returns

Model 2: Autoregressive Kalman Filter

We begin by extending the idea from Model 1. Instead of assuming PEP returns depend only on KO, we now assume they also depend on their own past value. This gives the observation model:

$$r_t^{PEP} = \alpha_t + \beta_1 r_{t-1}^{KO} + \beta_2 r_{t-1}^{PEP} + v_t$$

with Gaussian measurement noise:

$$v_t \sim \mathcal{N}(0, R)$$

Conditional probability model

This implies the likelihood:

$$p(r_t^{PEP} \mid \alpha_t, \beta_1, \beta_2) = \mathcal{N}\left(\alpha_t + \beta_1 r_{t-1}^{KO} + \beta_2 r_{t-1}^{PEP},\, R\right)$$

Assuming independence over time:

$$p(\mathbf{r}) = \prod_t p(r_t^{PEP} \mid x_t)$$

Log-likelihood and least squares form

Taking the log gives:

$$\log p(\mathbf{r}) = -\frac{1}{2} \sum_t \frac{\left(r_t^{PEP} - \alpha_t - \beta_1 r_{t-1}^{KO} - \beta_2 r_{t-1}^{PEP}\right)^2}{R} + C$$

Maximizing likelihood is therefore equivalent to minimizing:

$$\sum_t \left(r_t^{PEP} - \alpha_t - \beta_1 r_{t-1}^{KO} - \beta_2 r_{t-1}^{PEP}\right)^2$$

This shows the model is still a mean squared error minimizer under Gaussian noise.

Linear observation form

We rewrite the system in matrix form:

$$r_t^{PEP} = H_t x_t + v_t$$

where the state vector is:

$$x_t = \begin{bmatrix} \alpha_t \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

and the observation matrix is:

$$H_t = \begin{bmatrix} 1 & r_{t-1}^{KO} & r_{t-1}^{PEP} \end{bmatrix}$$

Why this becomes a Kalman filter problem

The system is now linear in the unknown parameters x_t, but these parameters are not fixed. To allow time variation, we assume a random walk:

$$x_t = x_{t-1} + w_t$$

$$w_t \sim \mathcal{N}(0, Q)$$

This assumption encodes that:

- α_t slowly drifts over time
- β_1 changes with market coupling strength
- β_2 changes with memory effects

Prediction step

Taking conditional expectation:

$$\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1}$$

Covariance propagation:

$$P_{t|t-1} = P_{t-1|t-1} + Q$$

Observation prediction and innovation

Predicted return:

$$\hat{r}_t^{PEP} = H_t \hat{x}_{t|t-1}$$

Innovation:

$$e_t = r_t^{PEP} - H_t \hat{x}_{t|t-1}$$

Innovation covariance:

$$S_t = H_t P_{t|t-1} H_t^T + R$$

Optimal update from least squares minimization

Minimizing posterior mean squared error gives:

$$K_t = P_{t|t-1} H_t^T S_t^{-1}$$

Update:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t e_t$$

Covariance update:

$$P_{t|t} = (I - K_t H_t) P_{t|t-1}$$

Final regression interpretation

Substituting back into the observation model yields:

$$r_t^{PEP} = \alpha_t + \beta_1 r_{t-1}^{KO} + \beta_2 r_{t-1}^{PEP}$$

Where:

- r_{t-1}^{KO}: cross-asset influence term
- r_{t-1}^{PEP}: autoregressive memory term
- β_1: coupling strength between assets
- β_2: persistence (momentum) effect
- α_t: time-varying drift

Model 3: Adaptive Volatility Kalman Filter

We now extend Model 2 by relaxing the assumption that uncertainty in the system is constant. Empirically, financial markets exhibit volatility clustering, so we allow the process noise to vary over time.

We start from the same structural model:

$$r_t^{PEP} = \alpha_t + \beta_1 r_{t-1}^{KO} + \beta_2 r_{t-1}^{PEP} + v_t$$

with:

$$v_t \sim \mathcal{N}(0, R)$$

Observation likelihood

As before, this implies:

$$p(r_t^{PEP} \mid x_t) = \mathcal{N}(H_t x_t, R)$$

where:

$$x_t = \begin{bmatrix} \alpha_t \\ \beta_1, t \\ \beta_2, t \end{bmatrix}, \quad H_t = \begin{bmatrix} 1 & r_{t-1}^{KO} & r_{t-1}^{PEP} \end{bmatrix}$$

State evolution with changing uncertainty

Unlike previous models, we now assume:

$$x_t = x_{t-1} + w_t$$

but the key difference is:

$$w_t \sim \mathcal{N}(0, Q_t)$$

This means the system is still a random walk, but the size of its uncertainty changes over time.

Volatility-driven process noise

We define rolling market volatility as:

$$\sigma_t^2 = \mathrm{Var}(r_{t-20:t}^{PEP})$$

and link it to model uncertainty:

$$Q_t = (c + \lambda \sigma_t^2) I$$

This ensures:

- low volatility $\Rightarrow$ stable parameters
- high volatility $\Rightarrow$ rapidly changing parameters

Prediction step

Expectation:

$$\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1}$$

Covariance propagation:

$$P_{t|t-1} = P_{t-1|t-1} + Q_t$$

Observation prediction

Predicted return:

$$\hat{r}_t^{PEP} = H_t \hat{x}_{t|t-1}$$

Innovation:

$$e_t = r_t^{PEP} - H_t \hat{x}_{t|t-1}$$

Innovation covariance

$$S_t = H_t P_{t|t-1} H_t^T + R$$

Kalman gain and update

Kalman gain:

$$K_t = P_{t|t-1} H_t^T S_t^{-1}$$

Update:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t e_t$$

$$P_{t|t} = (I - K_t H_t) P_{t|t-1}$$

Final regression interpretation

The predictive structure remains:

$$r_t^{PEP} = \alpha_t + \beta_1 r_{t-1}^{KO} + \beta_2 r_{t-1}^{PEP}$$

Where the key difference is that parameter uncertainty is now time-varying through:

$$Q_t = (c + \lambda \sigma_t^2) I$$

Where:

- Q_t: adaptive process noise controlling parameter flexibility
- σ_t^2: rolling market volatility
- c: baseline uncertainty level
- λ: sensitivity of the model to volatility changes
- I: identity matrix ensuring independent parameter diffusion

Evaluation Methodology

We first predict returns, then reconstruct prices:

$$\hat{P}_t = \hat{P}_{t-1}(1 + \hat{r}_{t-1})$$

This means:

- each prediction affects all future values
- small errors accumulate over time (compounding error)

We evaluate using:

- absolute error
- squared error
- cumulative error

Empirical Results

Figure 1: Kalman Filter vs Market Price

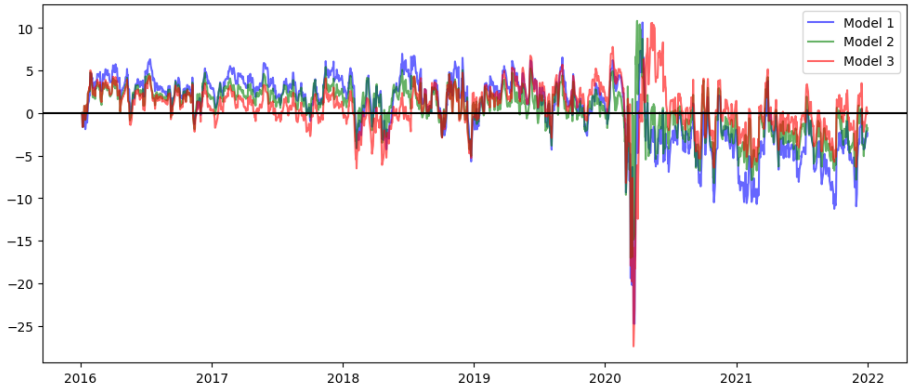

Figure 2: Residual Errors Over Time

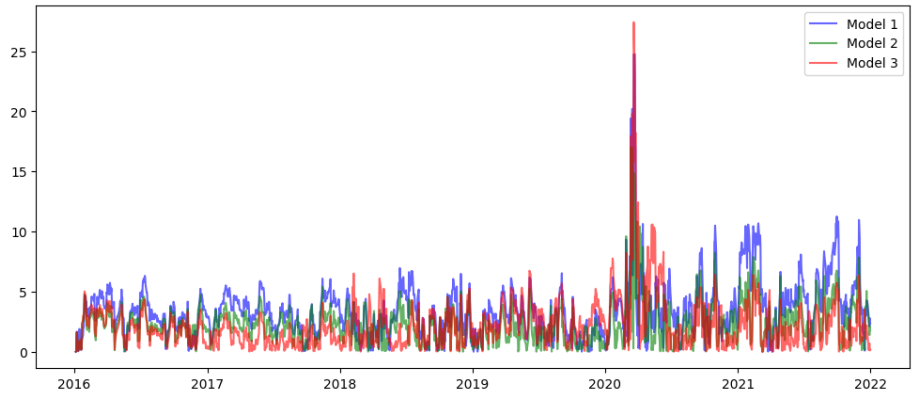

Figure 3: Absolute Errors

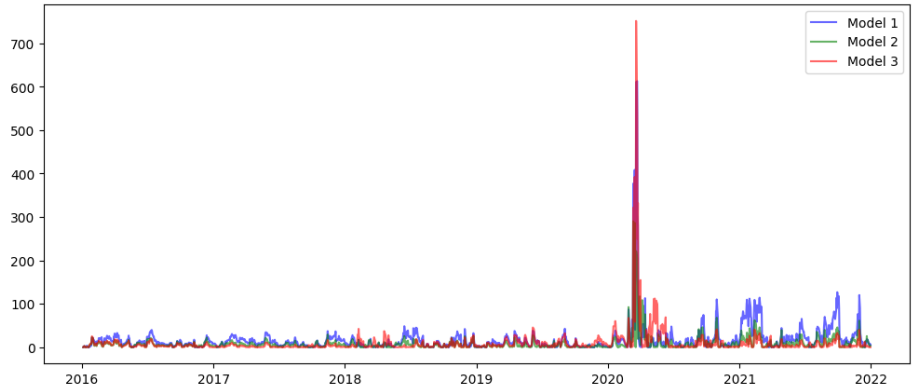

Figure 4: Squared Errors

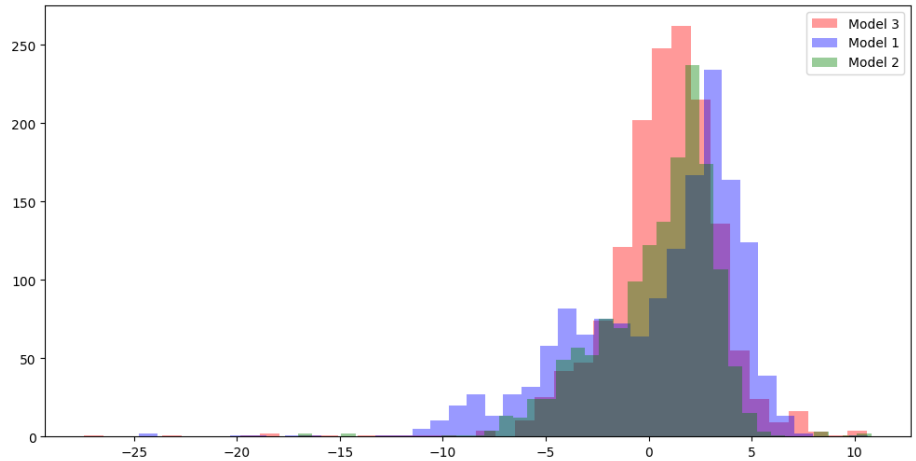

Figure 5: Error Distributions

Structural Breaks and Regime-Dependent Failure

Breakdown of Model Assumptions

All three models are derived under the same fundamental assumption:

$$x_t = x_{t-1} + w_t$$

which implies that the underlying relationship between assets evolves gradually over time with small stochastic perturbations.

Under this assumption, the system is locally smooth, and uncertainty grows in a controlled, approximately linear manner through the process noise.

However, during periods such as the COVID-19 shock in 2020, this assumption no longer holds. Empirically, the return-generating process exhibits abrupt and nonlinear changes:

$$P(r_t \mid \mathcal{F}_{t-1}) \neq P(r_{t+\Delta} \mid \mathcal{F}_{t-1})$$

This indicates a structural break in the data-generating process, where the statistical properties of returns change discontinuously rather than smoothly.

As a result, the Kalman filter prediction step:

$$\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1}$$

becomes systematically biased, since it assumes continuity where none exists.

Common Behaviour Across All Models

Across all three models, a consistent pattern is observed:

- In regimes where market noise is relatively stable and slowly varying, all models produce accurate predictions.
- In these regimes, residual errors remain small and approximately Gaussian.
- The mean squared error (MSE) grows gradually, consistent with the assumptions of the filter.

However, when the system undergoes rapid or nonlinear changes in volatility:

- the models systematically misestimate uncertainty,
- prediction errors increase sharply,
- and the filters tend to overreact to new observations.

This overreaction is a direct consequence of the recursive update:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t e_t$$

When the innovation e_t becomes large due to an unexpected regime shift, the update step introduces disproportionately large corrections, leading to instability.

Regime-Dependent Performance of Model 3

The adaptive volatility model (Model 3) highlights this effect most clearly.

In stable regimes, where volatility evolves slowly, the model achieves the best performance among all three models:

- lowest mean squared error (MSE)
- smallest residual variance
- most consistent tracking of returns

This is because the adaptive process noise:

$$Q_t = (c + \lambda \sigma_t^2) I$$

allows the model to appropriately scale its flexibility. When σ_t^2 is low, the system remains stable while still adapting to small changes in the underlying relationship.

However, during periods of rapidly increasing or highly nonlinear volatility:

- σ_t^2 rises sharply,
- Q_t becomes large,
- and the model allows excessive parameter variation.

This leads to:

- overestimation of uncertainty,
- amplification of noise rather than signal,
- and degradation of predictive accuracy.

As a result, Model 3 transitions from being the most accurate model in stable conditions to the least reliable during high-volatility regimes.

Interpretation

These results show that the limitation is not the Kalman filter framework itself, but the mismatch between its assumptions and the true structure of financial data.

All models implicitly assume:

- smooth parameter evolution,
- approximately Gaussian noise,
- and stable second-order statistics.

Financial markets violate these assumptions through:

- sudden regime shifts,
- nonlinear volatility growth,
- and heavy-tailed return distributions.

Thus, the filters perform well only when the market behaves approximately like the assumed model, and fail when this approximation breaks down.

Conclusion

The results show that Kalman filtering provides an effective framework for estimating time-varying relationships in financial data, but only under conditions where its assumptions are approximately satisfied.

In regimes with relatively stable and slowly varying noise, all models produce accurate predictions, with the adaptive volatility model achieving the best performance due to its ability to dynamically adjust uncertainty.

However, during periods of abrupt or nonlinear changes in market conditions, all models degrade in performance. In particular, the adaptive model, while optimal in stable regimes, becomes highly sensitive to volatility spikes and tends to overestimate uncertainty, leading to increased prediction error.

This demonstrates that Kalman filtering is fundamentally a locally valid approximation: it performs well when the system evolves smoothly, but fails under structural breaks and regime changes.

The primary limitation, therefore, is not the recursive estimation procedure itself, but the underlying assumptions of linearity, Gaussian noise, and smooth dynamics. Relaxing these assumptions — through nonlinear models, heavy-tailed noise distributions, or regime-switching frameworks — offers a promising direction for improving robustness in financial applications.

Overall, the Kalman filter remains a valuable baseline due to its interpretability and computational efficiency, but must be extended to handle the inherently unstable and nonlinear nature of financial markets.

References

- https://web.mit.edu/kirtley/kirtley/binlustuff/literature/control/Kalman%20filter.pdf
- Portions of the model construction and writing were done by GPT 5.3